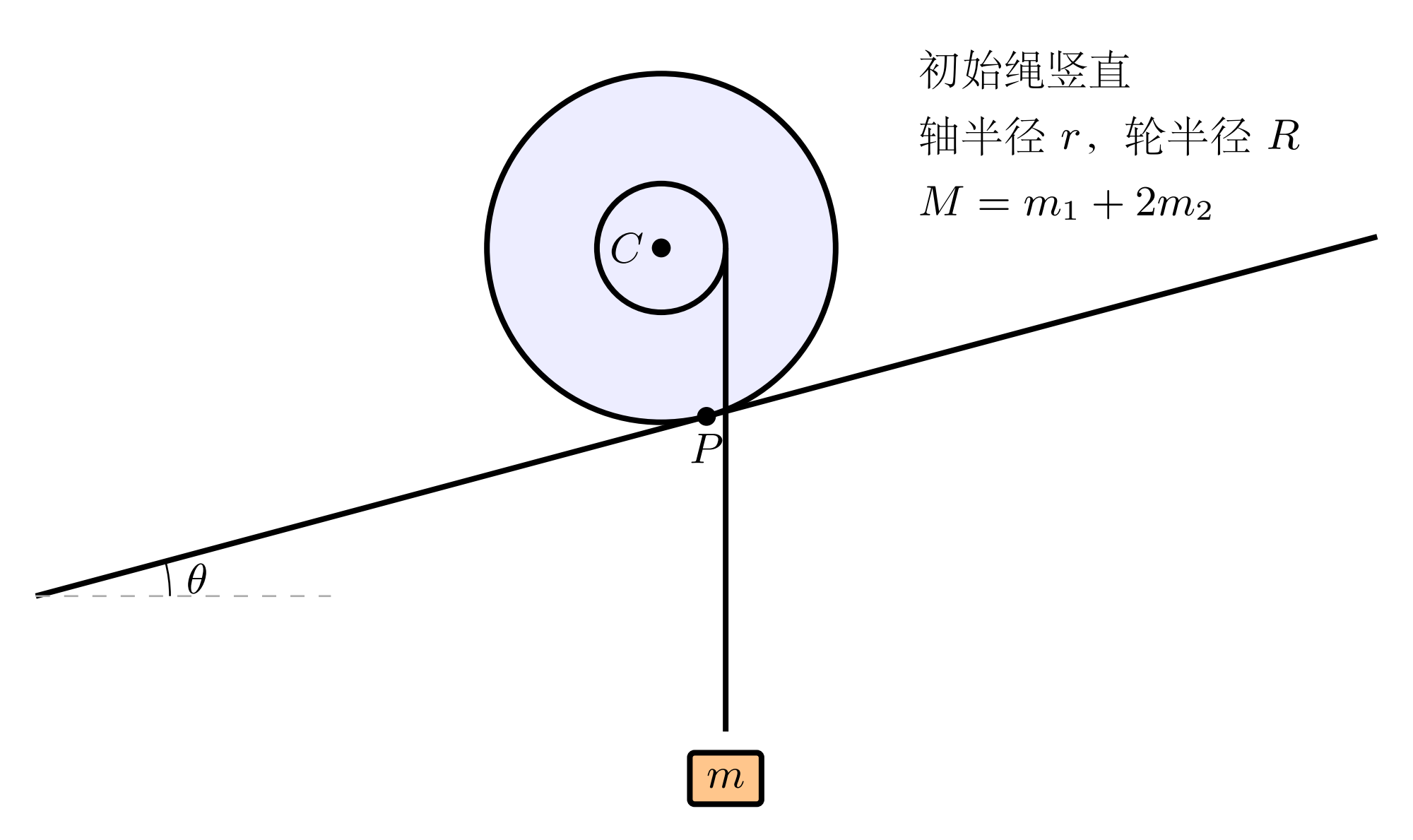
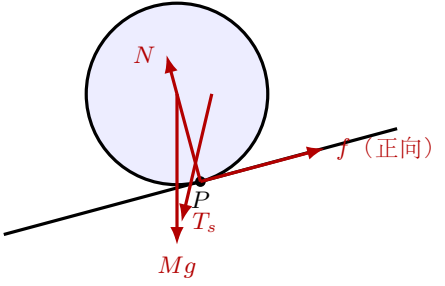
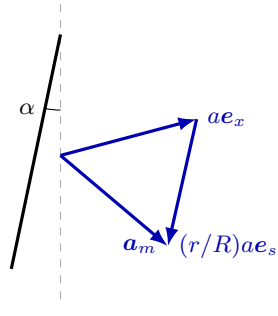
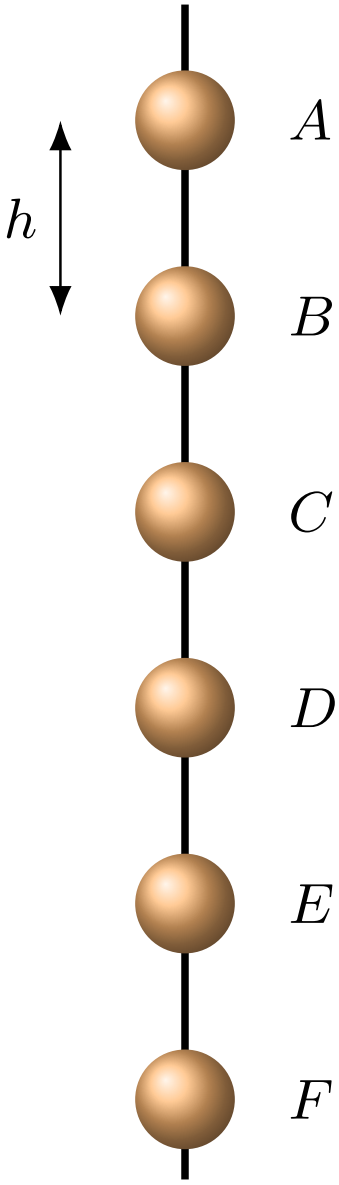

初始绳竖直
轴半径 r，轮半径 R
$M = m_1 + 2m_2$
C
P
θ
m



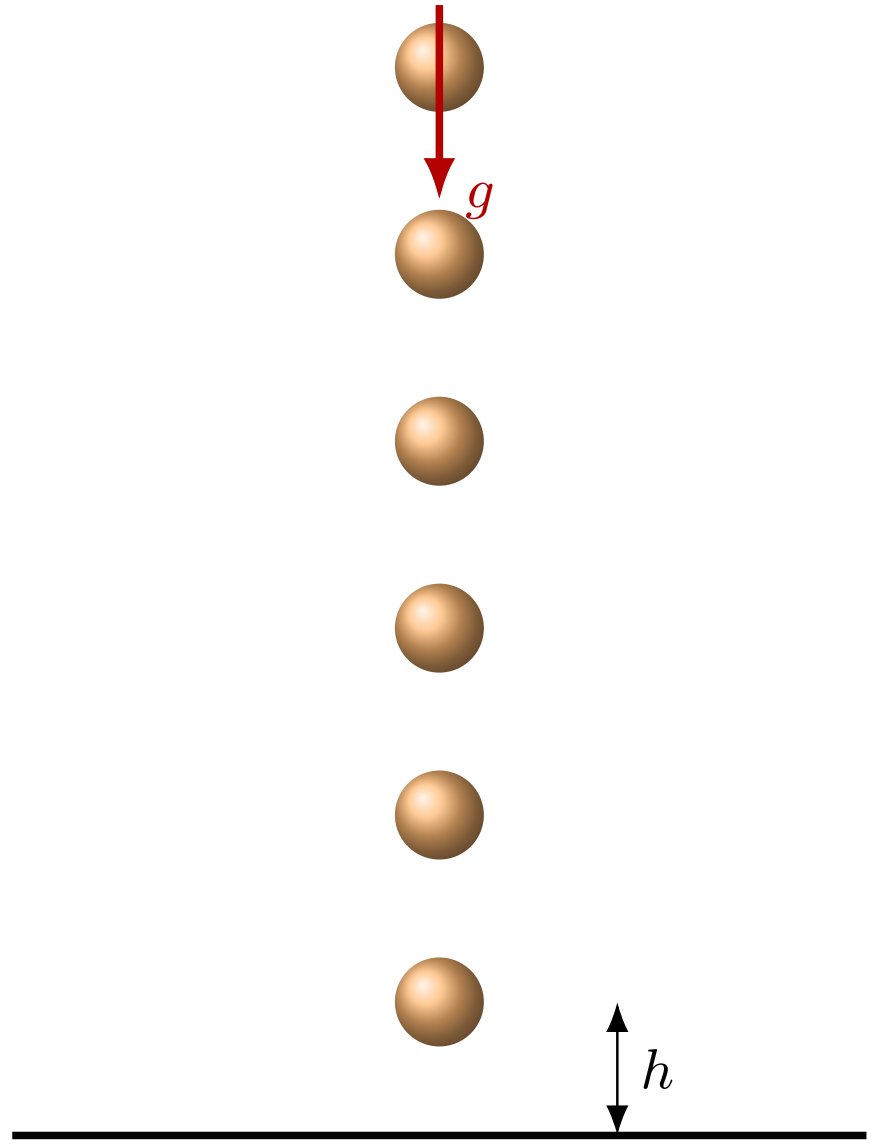
$$Ma = T_s[-\sin(\alpha+\theta)] + f - Mg\sin\theta$$



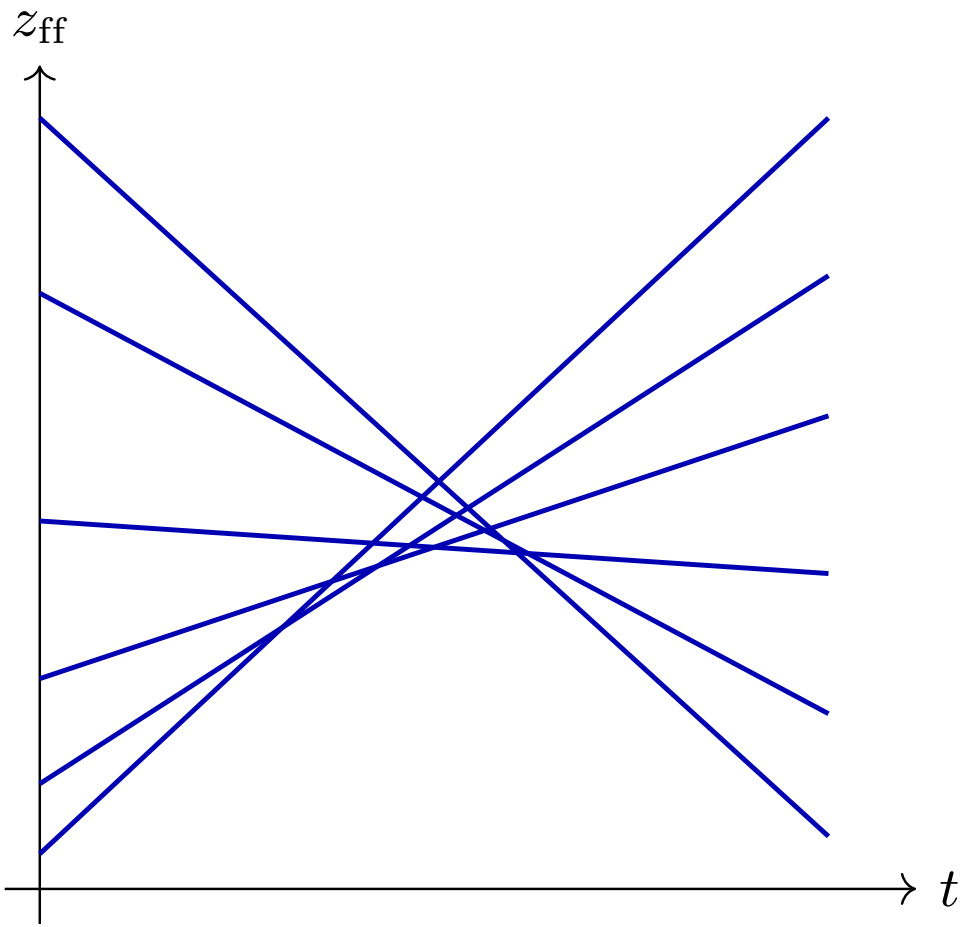
$$a\cos(\alpha+\theta) = g\sin\alpha$$



六球碰撞链

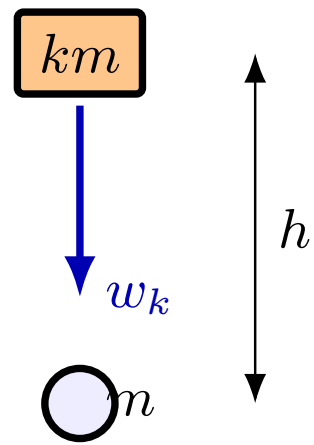


N 个质点的黏连坠落



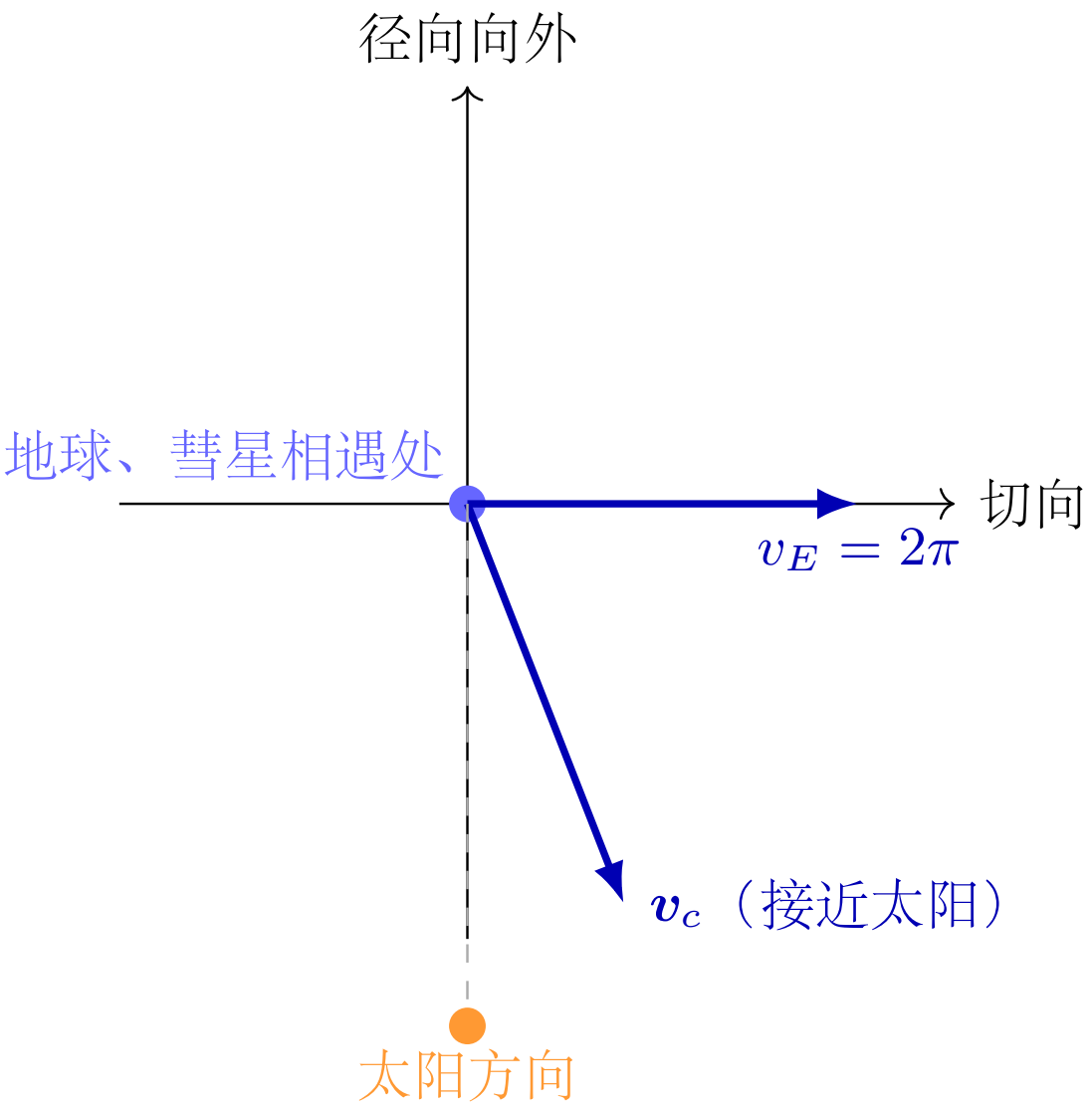
辅助轨迹穿越；实体球交换速度

最多 $\binom{6}{2} = 15$ 次



$$u_{k+1} = \frac{k}{k+1}\sqrt{u_k^2 + 2gh}$$

$$v_{\text{ground}}^2 = gh(N+1)(2N+1)/(3N)$$

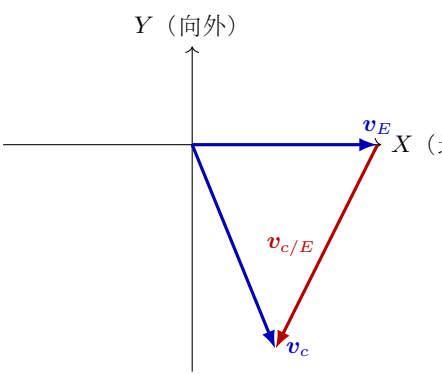


$T = 1200$ 年，$a = T^{2/3}$ AU

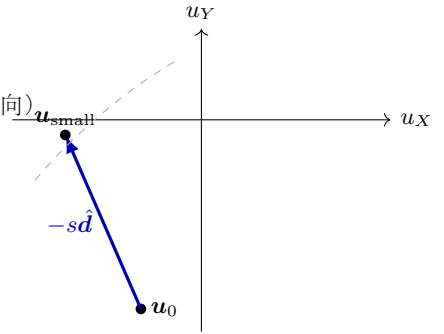
$r_1 = 0.1$ AU，$r_2 = 2a - r_1$

在 $r = 1$ AU 处取入境支

$v_t > 0, \quad v_r < 0$



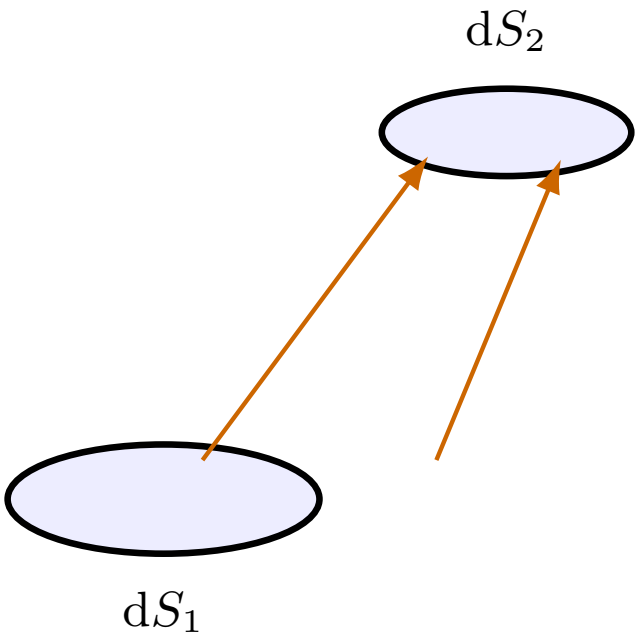
$v_{c/E} = 9.1$ AU/年



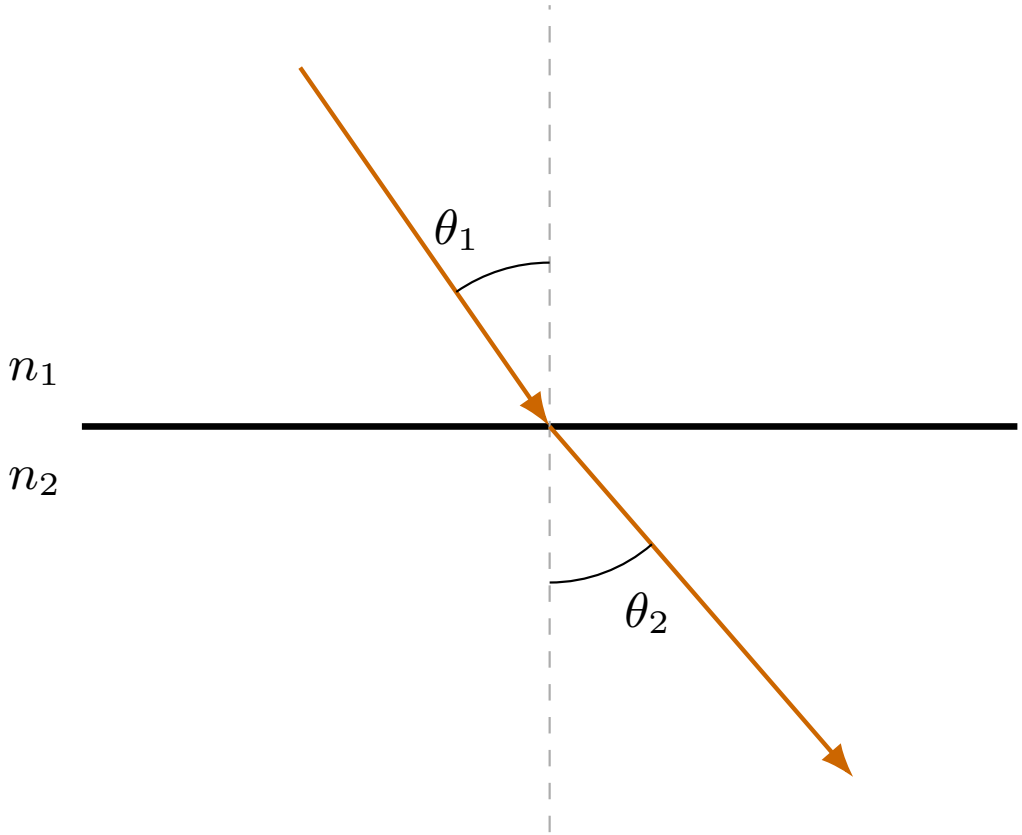
临界双曲线近地点 $r_p = R_E$

只沿彗星日心切向改变速度

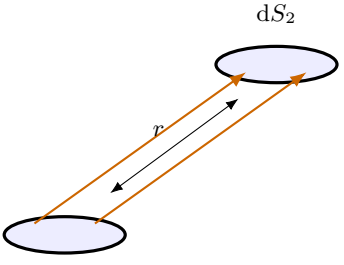
$s_{\min} = 4.0062 \times 10^4$ m/s



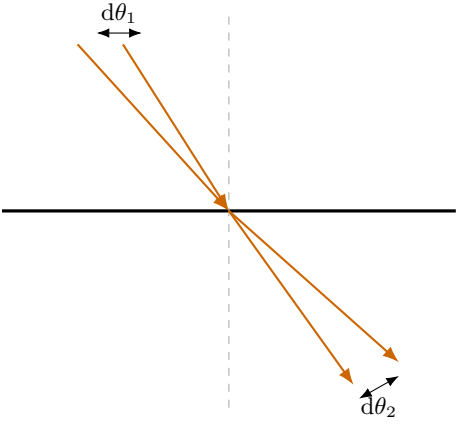
$$\mathrm{d}G = f(n)\,\mathrm{d}S_\perp\,\mathrm{d}\Omega$$



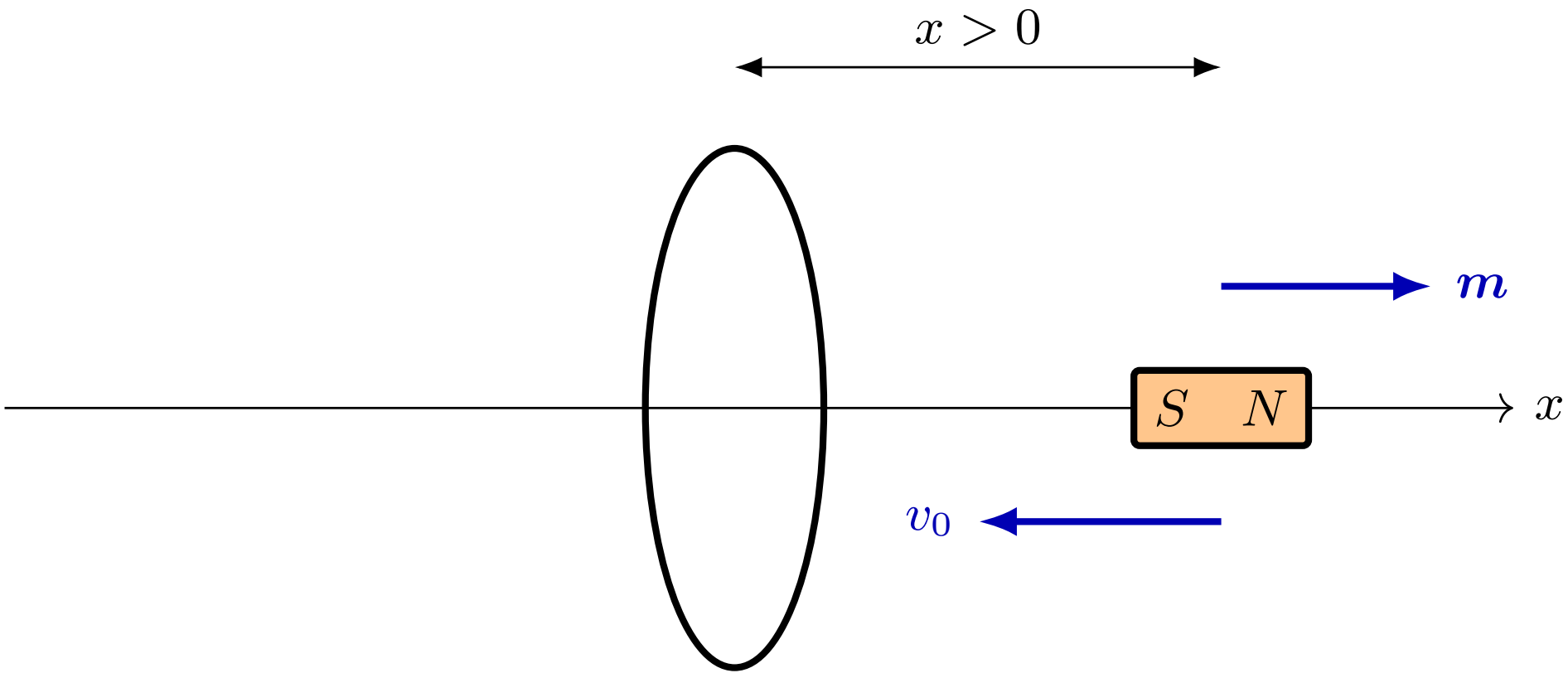
$$n_1 \sin\theta_1 = n_2 \sin\theta_2$$



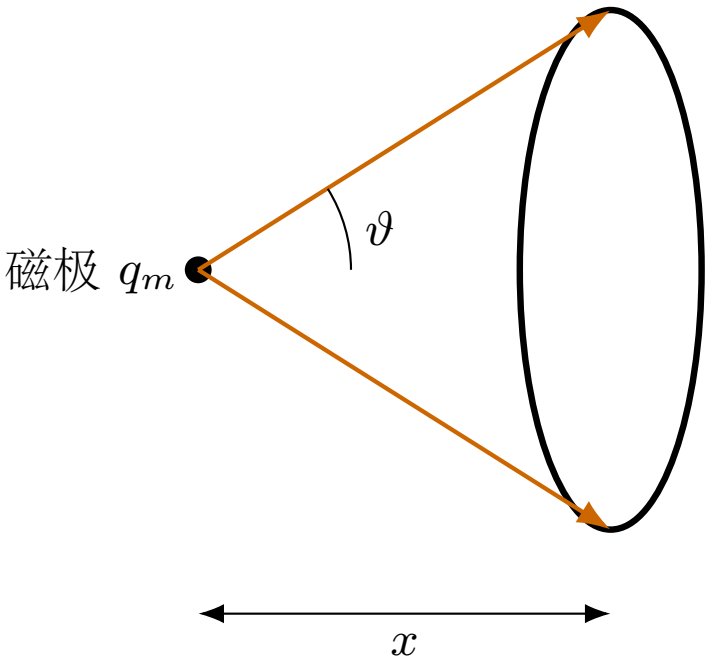
$$\mathrm{d}\Omega_1 = \frac{\mathrm{d}S_2 \cos\theta_2}{r^2}$$



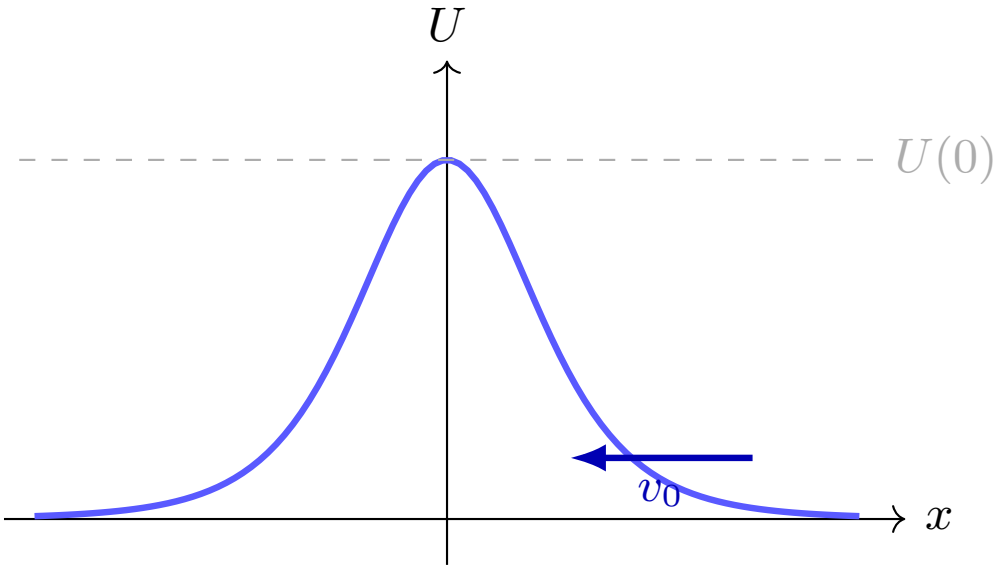
$$n_1 \cos\theta_1 \,\mathrm{d}\theta_1 = n_2 \cos\theta_2 \,\mathrm{d}\theta_2$$



闭合无电阻环，L

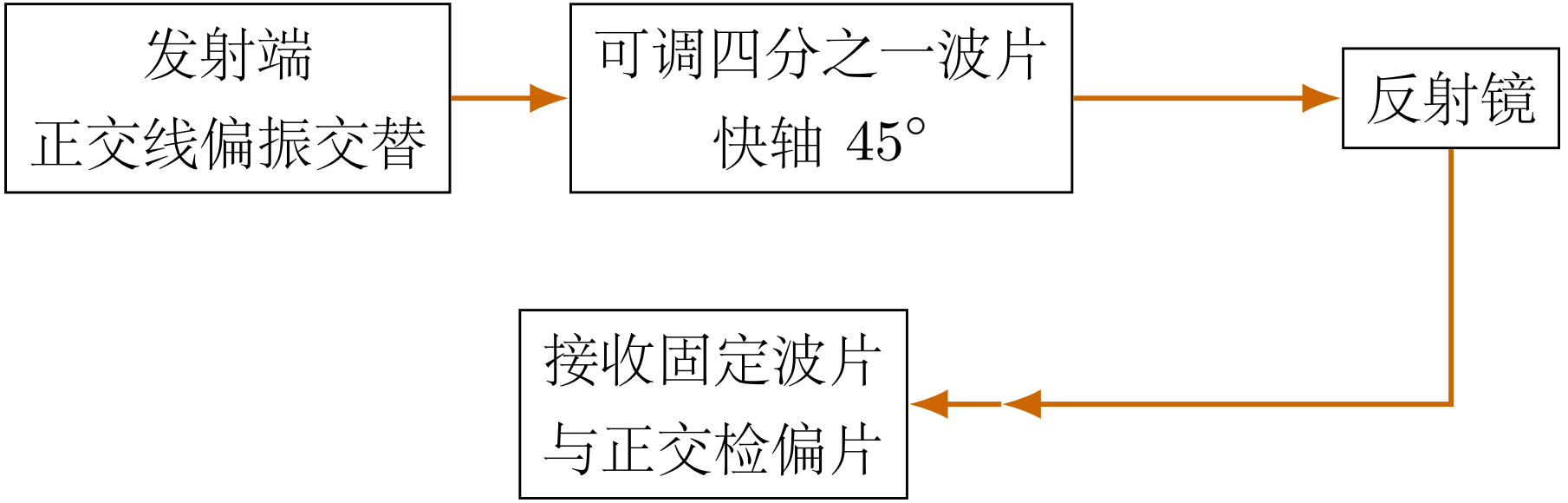


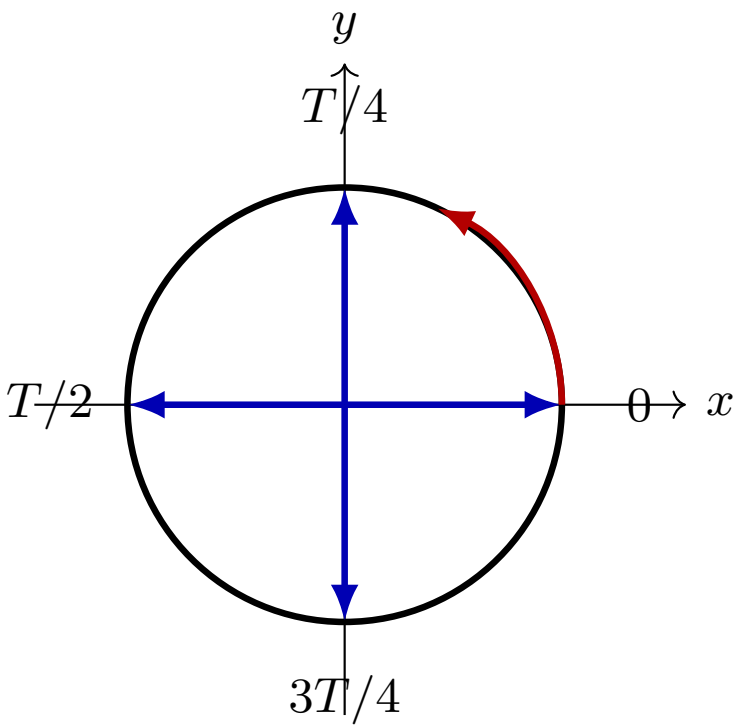
$$\Omega = 2\pi(1 - \cos\vartheta)$$



$$U(x) = \frac{\Phi^2(x)}{2L}$$

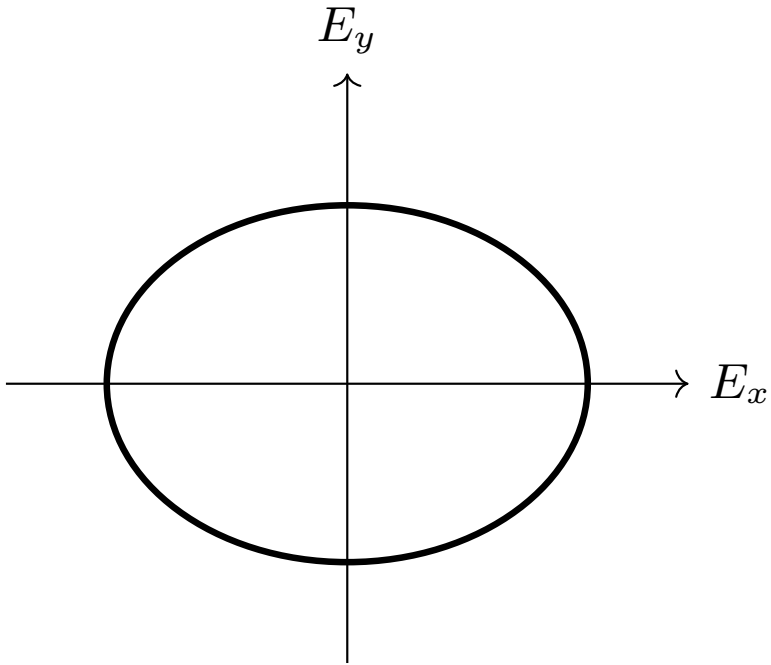
各色发射波片调到 $\delta = \pi/2$；接收波片固定





$e^{-i\omega t}$ 约定；电矢量随时间逆时针

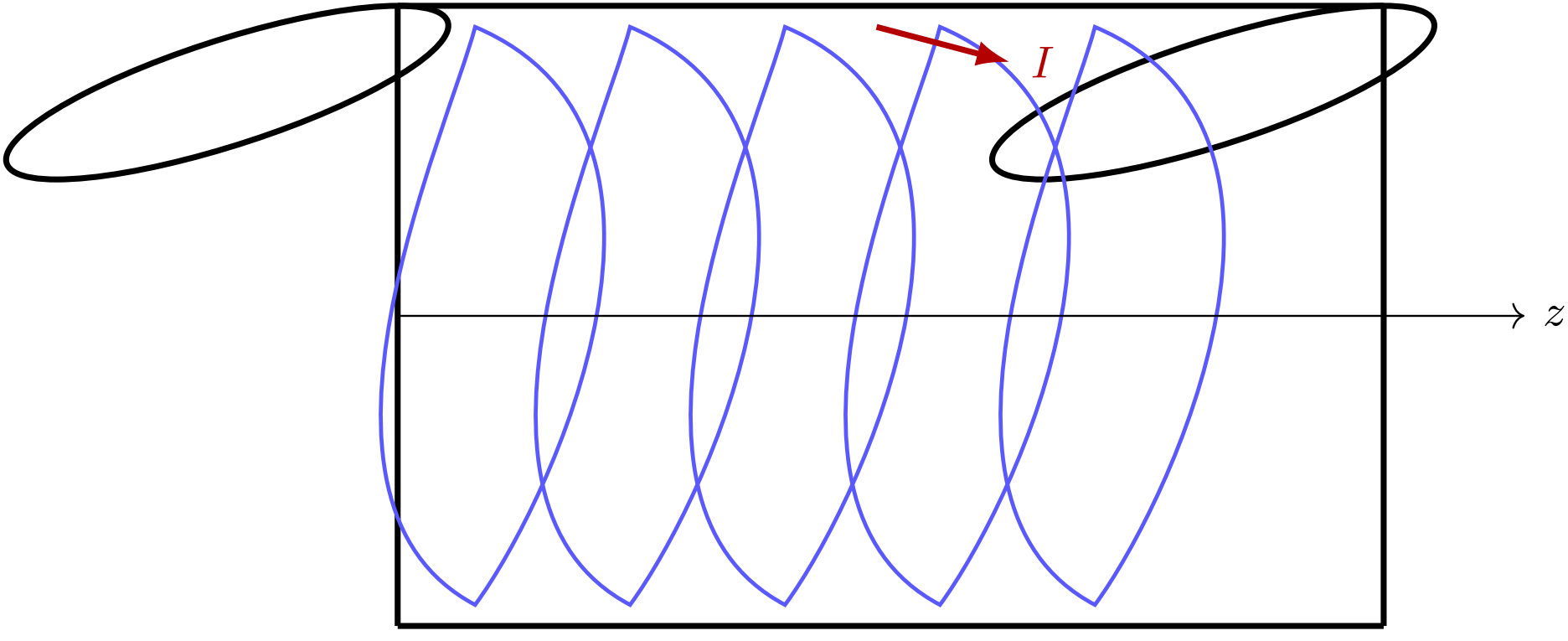
手性还须声明观察方向



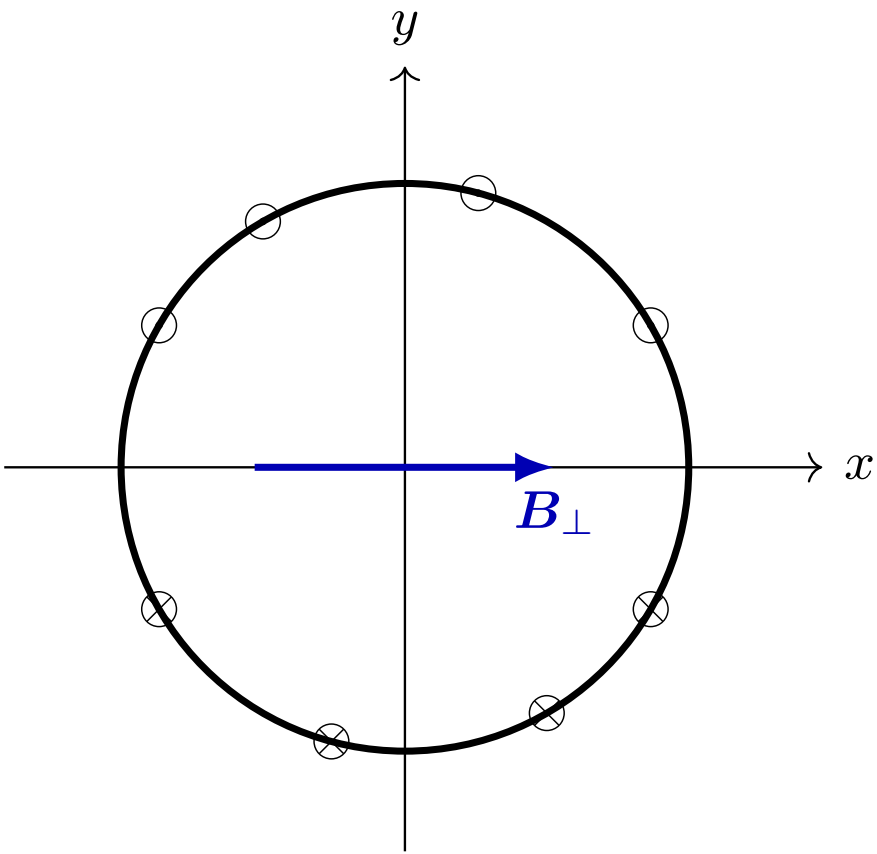
相位误差 $\varepsilon = \delta - \pi/2$

$I_{\text{ghost}}/I_{\text{normal}} = \tan^2(\varepsilon/2)$

小误差：$\simeq \varepsilon^2/4$

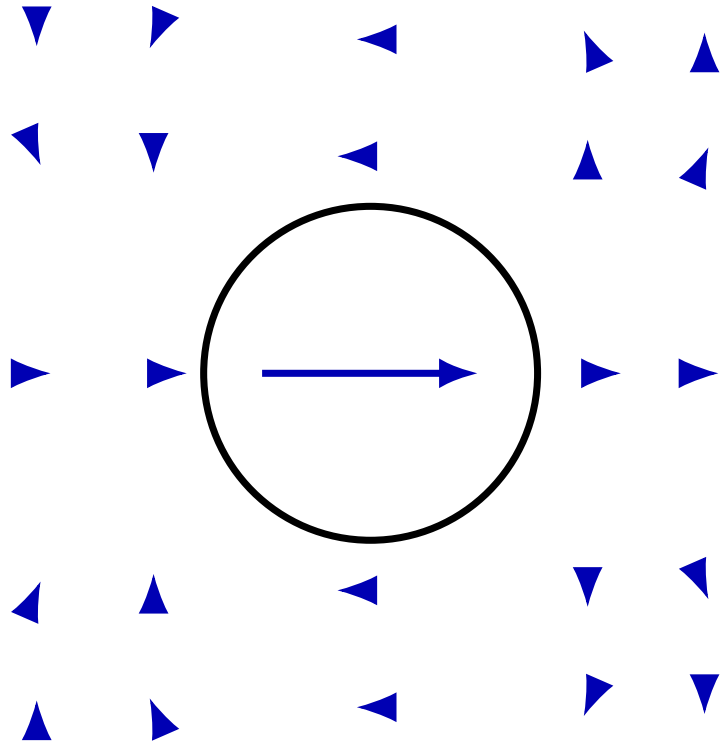


圆柱面上的倾斜椭圆匝



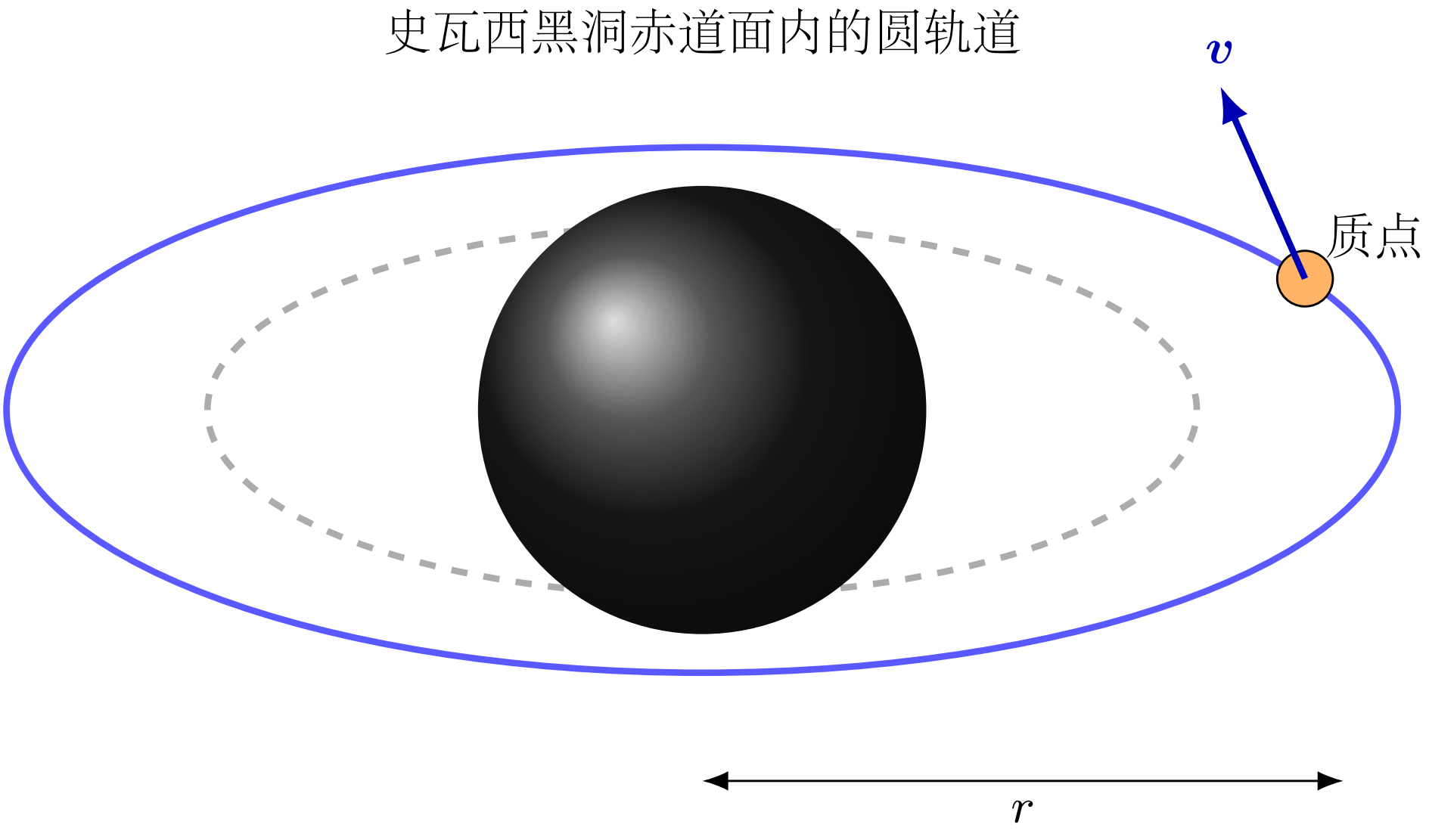
$K_z = nI\tan\theta\sin\varphi$

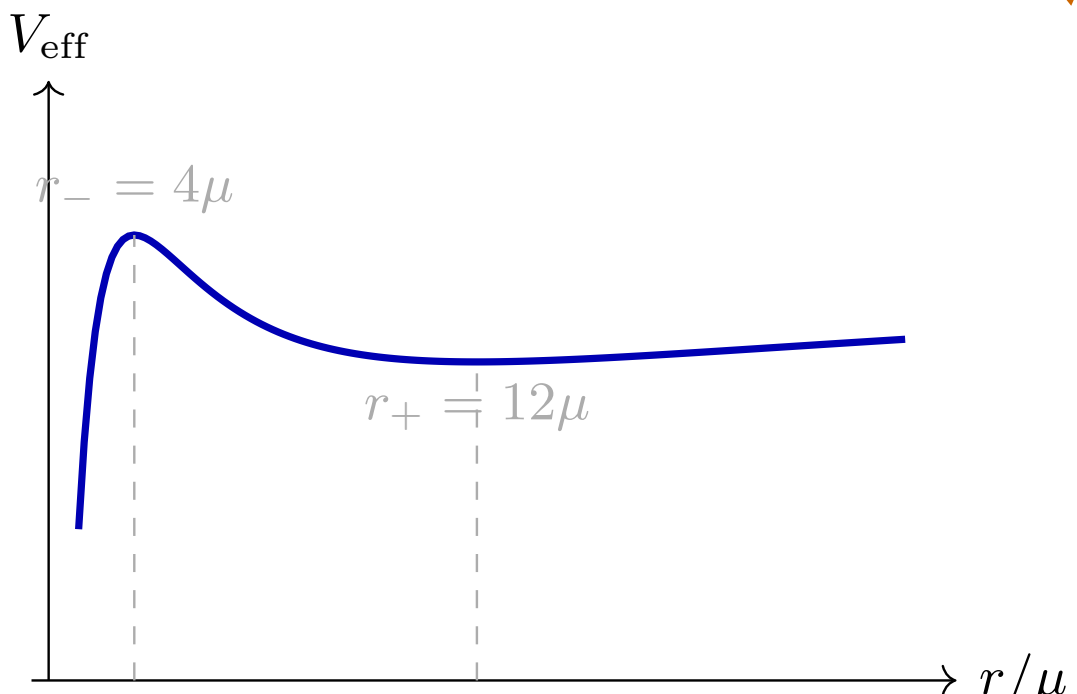
$\odot$：$+z$；$\otimes$：$-z$

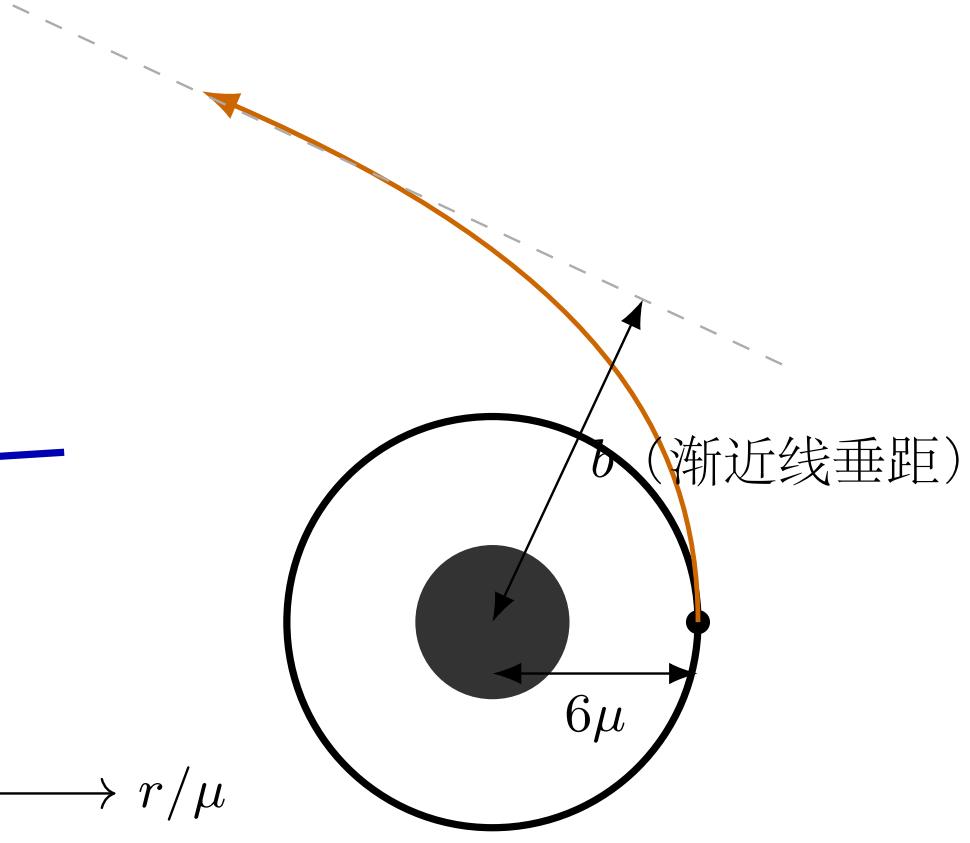


横截面外场按 ρ^{-2} 衰减

内部横场均匀；$B_z = \mu_0 nI$


史瓦西黑洞赤道面内的圆轨道
v
质点
r





示例 $L = 4GMm_0/c$；稳定支 $r_+ > 6\mu$

光路示意；$\mu = GM/c^2$

$b = 3\sqrt{6}\mu, \quad \nu_\infty = \sqrt{2}\nu$